ITCS347
Analysis and Design of Algorithms
Second Semester 2024/2025
First Midterm– One Hour

ID: __________ Section: ______ Serial: ______

Name: ______________________________

Question:	1	2	3	4	Total
Points:	7	8	7	8	30
Score:					

Instructor: Dr. Ali Alsaffar.

(1) Answer the following questions.

(a) **[2 points]** Suppose after analyzing your algorithm you found it was inefficient. Name one thing to do to improve the efficiency of your algorithm and in which step of the Algorithmic Problem Solving?

> **Solution:**
>
> 1. Consider making the algorithm approximate. Step 2: Decide on algorithm characteristics.
> 2. Consider applying different data structures. Step 2.
> 3. Transform the algorithm from sequential algorithm to parallel algorithm. Step 2.
> 4. Apply different design technique. Step 3: Design an Algorithm.
> 5. ...

(b) **[2 points]** Why time in seconds or milliseconds is not suitable to measure the time efficiency of algorithms? List two reasons.

> **Solution:**
>
> 1. Dependence on Computer speed.
> 2. Dependence on Quality of implementation (coding).
> 3. Dependence on the compiler generating the code.
> 4. Difficulty of clocking the actual running time.

(c) **[3 points]** Fill in the table below for each *iteration* of the **Selection Sort** algorithm when it is applied on the array: $A = [1, 7, 5, 0, 2]$.

Values of A

Iteration	1	7	5	0	2
1	0	7	5	1	2
2	0	1	5	7	2
3	0	1	2	7	5
4	0	1	2	5	7

(2) **[8 points]** Suppose you are given an integer C and an array of integers, $A[0..n-1]$ of size n, that is *sorted in ascending order*. Design a *linear* (single loop) and *in-place* algorithm that inserts C in the proper place in the array, keeping the array sorted. For illustration, below is an example.

Before inserting $C = 20$

$A \rightarrow$	2	5	9	12	17	25	26	31

After inserting $C = 20$

2	5	9	12	17	20	25	26	31

Solution: Shifting from the end.

Algorithm INSERT$(A[0..n-1], C)$
- $i \leftarrow n - 1$
- **while** $i \geq 0$ **and** $A[i] > C$ **do**
  - $A[i+1] \leftarrow A[i]$
  - $i \leftarrow i - 1$
- $A[i+1] \leftarrow C$

Using SWAP.

Algorithm INSERT$(A[0..n-1], C)$
- $A[n] \leftarrow C$
- $i \leftarrow n - 1$
- **while** $i \geq 0$ **and** $A[i] > C$ **do**
  - SWAP$(A[i+1], A[i])$
  - $i \leftarrow i - 1$

(3) Consider the following program segment.

1: $i \leftarrow 3, \quad p \leftarrow 1$
2: **while** $i \leq n + 2$ **do**
3:     **if** $i \bmod 4 \ = 0$ **then**
4:         $p \leftarrow p * i$
5:     $i \leftarrow i + 1$

(a) **[5 points]** Find number of times the multiplication $M(n)$ at line 4 is executed? Use summations to express your answer.

Solution: $M(n) = \displaystyle\sum_{i=4,8,\ldots,4k} 1 = \sum_{k=1}^{k} 1 = k.$

The condition of the loop $i \leq n+2$ in the last iteration is $4k \leq n + 2$.

Estimated answer: Assume $4k = n + 2 \implies \therefore k = (n+2)/4$

$\therefore M(n) = (n+2)/4$

Exact answer:

$$4k \leq n + 2 < 4(k+1)$$
$$k \leq (n+2)/4 < k + 1$$
$$\therefore \left\lfloor \frac{n+2}{4} \right\rfloor = k$$

$\therefore M(n) = \left\lfloor \dfrac{n+2}{4} \right\rfloor$

(b) **[2 points]** If the iterator i at line 5 is replaced by $i \leftarrow i + 2$, repeat part (a).

Solution: $M(n) = 0$.

(4) Consider the following recursive algorithm.

```
1: Algorithm F(n)
2:     if (n > 0) then
3:         s ← F(n − 1)
4:         for i ← 1 to 10 do
5:             s ← s + i
```

(a) **[2 points]** Set up a recurrence relation for the time complexity of the algorithm $C(n)$. What is the basic operation?

Solution: The basic operation is the addition at line 5.
The time complexity recurrence $T(n)$:

$$C(n) = \begin{cases} 0 & n = 0 \\ 1 + C(n-1) + 10 & n > 0 \end{cases}$$

$\textsc{Dir}(0) = 0, \textsc{Pre}(n) = 1$ subtraction$, \textsc{Post}(n) = 10$ additions
Note. In the post work, you may add $+10$ for the increment.

(b) **[5 points]** Find the solution of $C(n)$.

Solution:

$$\begin{aligned} CT(n) &= C(n-1) + 11 && \leftarrow \text{step } 1 \\ &= C(n-1) + 11 + 11 && \leftarrow \text{step } 2 \\ &= C(n-1) + 11 + 11 + 11 && \leftarrow \text{step } 3 \\ &\;\;\vdots \\ &= C(n-1) + 11k && \leftarrow \text{step } k \end{aligned}$$

Until $C(n-k) = C(0),\ n - k = 0 \implies k = n$, and
$\therefore C(n) = C(0) + 11n = 11n$.

(c) **[1 point]** Establish the order of growth of $C(n)$.

Solution:

$$C(n) \in \Theta(n)$$